COMPREHENSIVE ALGORITHMIC COMPENDIUM OF WRATTS-NET: A DUAL-BRANCH WAVELET-RESNET FRAMEWORK WITH CROSS-ATTENTION AND SUGENO FUZZY FUSION FOR NUCLEAR INSTANCE SEGMENTATION

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ABSTRACT

This compendium provides the complete, mathematically rigorous algorithmic specification of WRATTS-Net: a dual-branch neural architecture for nuclear instance segmentation in hematoxylin and eosin (H&E) histopathological images. WRATTS-Net integrates a Learnable Lifting Wavelet Transform (LLWT) stream for sub-pixel boundary preservation with a modified ResUNet stream for deep semantic context. Multi-scale channel-wise Bi-Directional Cross-Attention (BiCA) synchronizes representations across both streams without the quadratic spatial complexity of self-attention. ResUNet skip connections are regularized by a custom feature alignment loss ($\mathcal{L}_{\text{feat}}$) to prevent semantic divergence. Predictions from both streams are fused pixel-wise via a learnable Sugeno fuzzy integral with singleton densities (g_1, g_2). A multi-task HoVer distance regression head predicts horizontal and vertical distance-to-centroid maps, driving marker-controlled watershed separation of touching cells. Every mathematical operation, tensor transformation, loss evaluation, sliding-window inference procedure, post-processing heuristic, validation search, and benchmarking metric is formulated into formal, self-contained, publication-grade pseudocode algorithms formatted across full-width double-column spans.

***Index Terms*—** Computational Pathology, Nuclear Instance Segmentation, Learnable Lifting Wavelets, Sugeno Fuzzy Integral, Bi-Directional Cross-Attention, HoVer-Net, Multi-Task Learning.

1. OVERVIEW OF ALGORITHMIC MODULES

The proposed WRATTS-Net framework integrates 15 interconnected algorithmic procedures structured across six computational phases:

(i) **Global Model Architecture**: End-to-end forward inference (Algorithm 1, continued in Algorithm 1) and multi-scale ASPP bottleneck processing (Algorithm 5).

(ii) **Wavelet Multi-Resolution Processing**: Polyphase Learnable Lifting Wavelet Transform forward decomposition with stain-adaptive calibration (Algorithm 2) and symmetric lossless inverse synthesis (Algorithm 3).

(iii) **Feature Interactivity and Gating**: Generalized Mean (GeM) downsampling (Algorithm 4), Bi-Directional Channel Cross-Attention with identity initialization (Algorithm 6), and Skip-Connection Feature Alignment (Algorithm 7).

(iv) **Fuzzy Fusion and Supervision**: Learnable Sugeno Fuzzy Decision Fusion (Algorithm 8) and Multi-Task Composite Loss Evaluation (Algorithm 9).

(v) **Optimization and Inference**: End-to-end model training with AdamW, EMA shadow weights, and OneCycleLR (Algorithm 10), followed by raised-cosine sliding-window tile stitching (Algorithm 11).

(vi) **Instance Decoding and Benchmarking**: Marker-controlled distance-guided watershed decoding (Algorithm 12), validation hyperparameter grid search (Algorithm 13), Aggregated Jaccard Index (Algorithm 14), and Panoptic Quality metric evaluation (Algorithm 15).

2. GLOBAL ARCHITECTURE AND WAVELET FORMULATIONS

This section details the end-to-end inference pass and the forward and inverse mathematical formulations of the learnable lifting wavelet transform. To maintain visual clarity across publication pages, the complete end-to-end model pipeline is presented across Algorithm 1 (Part 1: Dual-Stream Hierarchical Encoder and ASPP Bottlenecks) and Algorithm 1 (Part 2: Symmetrical Decoders, Skip Feature Alignment, and Multi-Task Prediction Heads).

Algorithm 1 WRATTS-Net End-to-End Forward Pass (Part 1: Dual-Stream Hierarchical Encoder and ASPP Bottlenecks)

Require: Input RGB image patch $\mathbf{X} \in \mathbb{R}^{B\times 3\times H\times W}$

Ensure: Intermediate multi-scale encoder skip connections $\{\mathbf{s}_l^w, \mathbf{s}_l^r\}_{l=1}^4$, Wavelet detail bypass tensors $\{\mathbf{D}_l\}_{l=1}^4$, Calibrated bottleneck representations $\mathbf{x}_w, \mathbf{x}_r \in \mathbb{R}^{B\times 1024\times \frac{H}{16}\times \frac{W}{16}}$

// Stage 1: Dual-Branch 4-Stage Hierarchical Contracting Pathway

// Stage 1 ($H \times W \to \frac{H}{2} \times \frac{W}{2}$, **channels** $3 \to 64 \to 128$)**:**

$\mathbf{s}_1^w, \mathbf{x}_w, \mathbf{D}_1 \leftarrow \text{Enc1}_w(\mathbf{X})$ ▷ 3 ResBlocks, LLWT downsampling; $\mathbf{s}_1^w \in \mathbb{R}^{B\times 64\times H\times W}$, $\mathbf{D}_1 \in \mathbb{R}^{B\times 192\times \frac{H}{2}\times \frac{W}{2}}$

$\mathbf{s}_1^r, \mathbf{x}_r \leftarrow \text{Enc1}_r(\mathbf{X})$ ▷ 3 ResBlocks, GeM downsampling; $\mathbf{s}_1^r \in \mathbb{R}^{B\times 64\times H\times W}$, $\mathbf{x}_r \in \mathbb{R}^{B\times 128\times \frac{H}{2}\times \frac{W}{2}}$

$\mathbf{x}_w, \mathbf{x}_r \leftarrow \text{BiCA}_{e1}(\mathbf{x}_w, \mathbf{x}_r)$ ▷ Bi-Directional Channel Cross-Attention at channel width $C = 128$

// Stage 2 ($\frac{H}{2} \times \frac{W}{2} \to \frac{H}{4} \times \frac{W}{4}$, **channels** $128 \to 128 \to 256$)**:**

$\mathbf{s}_2^w, \mathbf{x}_w, \mathbf{D}_2 \leftarrow \text{Enc2}_w(\mathbf{x}_w)$ ▷ 3 ResBlocks, LLWT downsampling; $\mathbf{s}_2^w \in \mathbb{R}^{B\times 128\times \frac{H}{2}\times \frac{W}{2}}$, $\mathbf{D}_2 \in \mathbb{R}^{B\times 384\times \frac{H}{4}\times \frac{W}{4}}$

$\mathbf{s}_2^r, \mathbf{x}_r \leftarrow \text{Enc2}_r(\mathbf{x}_r)$ ▷ 3 ResBlocks, GeM downsampling; $\mathbf{s}_2^r \in \mathbb{R}^{B\times 128\times \frac{H}{2}\times \frac{W}{2}}$, $\mathbf{x}_r \in \mathbb{R}^{B\times 256\times \frac{H}{4}\times \frac{W}{4}}$

$\mathbf{x}_w, \mathbf{x}_r \leftarrow \text{BiCA}_{e2}(\mathbf{x}_w, \mathbf{x}_r)$ ▷ Bi-Directional Channel Cross-Attention at channel width $C = 256$

// Stage 3 ($\frac{H}{4} \times \frac{W}{4} \to \frac{H}{8} \times \frac{W}{8}$, **channels** $256 \to 256 \to 512$)**:**

$\mathbf{s}_3^w, \mathbf{x}_w, \mathbf{D}_3 \leftarrow \text{Enc3}_w(\mathbf{x}_w)$ ▷ 4 ResBlocks, LLWT downsampling; $\mathbf{s}_3^w \in \mathbb{R}^{B\times 256\times \frac{H}{4}\times \frac{W}{4}}$, $\mathbf{D}_3 \in \mathbb{R}^{B\times 768\times \frac{H}{8}\times \frac{W}{8}}$

$\mathbf{s}_3^r, \mathbf{x}_r \leftarrow \text{Enc3}_r(\mathbf{x}_r)$ ▷ 4 ResBlocks, GeM downsampling; $\mathbf{s}_3^r \in \mathbb{R}^{B\times 256\times \frac{H}{4}\times \frac{W}{4}}$, $\mathbf{x}_r \in \mathbb{R}^{B\times 512\times \frac{H}{8}\times \frac{W}{8}}$

$\mathbf{x}_w, \mathbf{x}_r \leftarrow \text{BiCA}_{e3}(\mathbf{x}_w, \mathbf{x}_r)$ ▷ Bi-Directional Channel Cross-Attention at channel width $C = 512$

// Stage 4 ($\frac{H}{8} \times \frac{W}{8} \to \frac{H}{16} \times \frac{W}{16}$, **channels** $512 \to 512 \to 1024$)**:**

$\mathbf{s}_4^w, \mathbf{x}_w, \mathbf{D}_4 \leftarrow \text{Enc4}_w(\mathbf{x}_w)$ ▷ 3 ResBlocks, LLWT downsampling; $\mathbf{s}_4^w \in \mathbb{R}^{B\times 512\times \frac{H}{8}\times \frac{W}{8}}$, $\mathbf{D}_4 \in \mathbb{R}^{B\times 1536\times \frac{H}{16}\times \frac{W}{16}}$

$\mathbf{s}_4^r, \mathbf{x}_r \leftarrow \text{Enc4}_r(\mathbf{x}_r)$ ▷ 3 ResBlocks, GeM downsampling; $\mathbf{s}_4^r \in \mathbb{R}^{B\times 512\times \frac{H}{8}\times \frac{W}{8}}$, $\mathbf{x}_r \in \mathbb{R}^{B\times 1024\times \frac{H}{16}\times \frac{W}{16}}$

$\mathbf{x}_w, \mathbf{x}_r \leftarrow \text{BiCA}_{e4}(\mathbf{x}_w, \mathbf{x}_r)$ ▷ Bi-Directional Channel Cross-Attention at channel width $C = 1024$

// Stage 2: Dual ASPP Bottlenecks and Bottleneck Cross-Attention

$\mathbf{x}_w \leftarrow \text{ASPPBottleneck}_w(\mathbf{x}_w)$ ▷ Rates $\{1, 4, 7\}$, depthwise-separable dilated convolutions, $C = 1024$

$\mathbf{x}_r \leftarrow \text{ASPPBottleneck}_r(\mathbf{x}_r)$ ▷ Rates $\{1, 4, 7\}$, depthwise-separable dilated convolutions, $C = 1024$

$\mathbf{x}_w, \mathbf{x}_r \leftarrow \text{BiCA}_{\text{bot}}(\mathbf{x}_w, \mathbf{x}_r)$ ▷ Deepest cross-stream representation calibration at $C = 1024$

return $\{\mathbf{s}_1^w, \ldots, \mathbf{s}_4^w, \mathbf{D}_1, \ldots, \mathbf{D}_4, \mathbf{s}_1^r, \ldots, \mathbf{s}_4^r, \mathbf{x}_w, \mathbf{x}_r\}$

Algorithm 1 WRATTS-Net End-to-End Forward Pass (Part 2: Symmetrical Decoders, Feature Alignment, and Heads, Continued)

Require: Encoder skips $\{\mathbf{s}_l^w, \mathbf{s}_l^r\}_{l=1}^4$, Detail bypasses $\{\mathbf{D}_l\}_{l=1}^4$, Calibrated bottleneck features $\mathbf{x}_w, \mathbf{x}_r \in \mathbb{R}^{B\times 1024\times \frac{H}{16}\times \frac{W}{16}}$
Ensure: Fused segmentation logits $\mathbf{z}_{\text{fused}} \in \mathbb{R}^{B\times 1\times H\times W}$,
Horizontal and vertical distance-to-centroid maps $\hat{\mathbf{H}}, \hat{\mathbf{V}} \in [-1,1]^{B\times 1\times H\times W}$,
Skip-connection feature alignment loss $\mathcal{L}_{\text{feat}} \in \mathbb{R}$,
Deep supervision auxiliary segmentation logits $\mathbf{z}_{\text{aux3}}, \mathbf{z}_{\text{aux2}}$ (during training)

// Stage 3: Symmetrical Multi-Stage Decoders with Inter-Branch BiCA

// Decoder Stage 4 ($\frac{H}{16}\times\frac{W}{16} \to \frac{H}{8}\times\frac{W}{8}$, **channels** $1024 \to 512$)**:**
$\mathbf{d}_4^w \leftarrow \text{Dec4}_w(\mathbf{x}_w, \mathbf{s}_4^w, \mathbf{D}_4)$ ▷ ILLWT detail synthesis + bilinear upsample + 3 Pre-activation ResBlocks
$\mathbf{d}_4^r, \mathbf{u}_4^r \leftarrow \text{Dec4}_r(\mathbf{x}_r, \mathbf{s}_4^r)$ ▷ 2×2 ConvTranspose2d upsample $\mathbf{u}_4^r$ + skip concat + 3 ResBlocks
$\mathbf{d}_4^w, \mathbf{d}_4^r \leftarrow \text{BiCA}_{d4}(\mathbf{d}_4^w, \mathbf{d}_4^r)$ ▷ Bi-Directional Channel Cross-Attention at channel width $C = 512$

// Decoder Stage 3 ($\frac{H}{8}\times\frac{W}{8} \to \frac{H}{4}\times\frac{W}{4}$, **channels** $512 \to 256$)**:**
$\mathbf{d}_3^w \leftarrow \text{Dec3}_w(\mathbf{d}_4^w, \mathbf{s}_3^w, \mathbf{D}_3)$ ▷ ILLWT detail synthesis + bilinear upsample + 4 Pre-activation ResBlocks
$\mathbf{d}_3^r, \mathbf{u}_3^r \leftarrow \text{Dec3}_r(\mathbf{d}_4^r, \mathbf{s}_3^r)$ ▷ 2×2 ConvTranspose2d upsample $\mathbf{u}_3^r$ + skip concat + 4 ResBlocks
$\mathbf{d}_3^w, \mathbf{d}_3^r \leftarrow \text{BiCA}_{d3}(\mathbf{d}_3^w, \mathbf{d}_3^r)$ ▷ Bi-Directional Channel Cross-Attention at channel width $C = 256$

// Decoder Stage 2 ($\frac{H}{4}\times\frac{W}{4} \to \frac{H}{2}\times\frac{W}{2}$, **channels** $256 \to 128$)**:**
$\mathbf{d}_2^w \leftarrow \text{Dec2}_w(\mathbf{d}_3^w, \mathbf{s}_2^w, \mathbf{D}_2)$ ▷ ILLWT detail synthesis + bilinear upsample + 3 Pre-activation ResBlocks
$\mathbf{d}_2^r, \mathbf{u}_2^r \leftarrow \text{Dec2}_r(\mathbf{d}_3^r, \mathbf{s}_2^r)$ ▷ 2×2 ConvTranspose2d upsample $\mathbf{u}_2^r$ + skip concat + 3 ResBlocks
$\mathbf{d}_2^w, \mathbf{d}_2^r \leftarrow \text{BiCA}_{d2}(\mathbf{d}_2^w, \mathbf{d}_2^r)$ ▷ Bi-Directional Channel Cross-Attention at channel width $C = 128$

// Decoder Stage 1 ($\frac{H}{2}\times\frac{W}{2} \to H\times W$, **channels** $128 \to 64$)**:**
$\mathbf{d}_1^w \leftarrow \text{Dec1}_w(\mathbf{d}_2^w, \mathbf{s}_1^w, \mathbf{D}_1)$ ▷ ILLWT detail synthesis + bilinear upsample + 3 Pre-activation ResBlocks
$\mathbf{d}_1^r, \mathbf{u}_1^r \leftarrow \text{Dec1}_r(\mathbf{d}_2^r, \mathbf{s}_1^r)$ ▷ 2×2 ConvTranspose2d upsample $\mathbf{u}_1^r$ + skip concat + 3 ResBlocks
$\mathbf{d}_1^w, \mathbf{d}_1^r \leftarrow \text{BiCA}_{d1}(\mathbf{d}_1^w, \mathbf{d}_1^r)$ ▷ Bi-Directional Channel Cross-Attention at channel width $C = 64$

// Stage 4: Skip-Connection Feature Alignment Loss

$\mathcal{L}_{\text{feat}} \leftarrow \sum_{l=1}^4 \left\| \frac{\mathbf{u}_l^r}{\|\mathbf{u}_l^r\|_2+\epsilon} - \frac{\mathbf{s}_l^r}{\|\mathbf{s}_l^r\|_2+\epsilon} \right\|_2^2$ ▷ Computed per Algorithm 7

// Stage 5: Multi-Task Prediction Heads and Sugeno Fuzzy Fusion

$\mathbf{z}_w \leftarrow \text{Head}_w(\mathbf{d}_1^w), \quad \mathbf{z}_r \leftarrow \text{Head}_r(\mathbf{d}_1^r)$ ▷ Independent branch logits ($64 \to 32 \to 1$)
$p_w \leftarrow \sigma(\mathbf{z}_w), \quad p_r \leftarrow \sigma(\mathbf{z}_r)$ ▷ Probability maps in range $[0,1]$
$p_{\text{fused}}, \mathbf{z}_{\text{fused}} \leftarrow \text{SugenoFusion}(p_w, p_r)$ ▷ Fuzzy integral decision fusion per Algorithm 8

// Deep Supervision Auxiliary Heads (active during training):

$\mathbf{z}_{\text{aux3}} \leftarrow \text{BilinearInterpolate}\left(\text{AuxHead3}([\mathbf{d}_3^w \,\|\, \mathbf{d}_3^r]),\ \text{size} = (H, W)\right)$ ▷ $[\mathbf{d}_3^w \,\|\, \mathbf{d}_3^r] \in \mathbb{R}^{B\times 512\times \frac{H}{4}\times \frac{W}{4}}$
$\mathbf{z}_{\text{aux2}} \leftarrow \text{BilinearInterpolate}\left(\text{AuxHead2}([\mathbf{d}_2^w \,\|\, \mathbf{d}_2^r]),\ \text{size} = (H, W)\right)$ ▷ $[\mathbf{d}_2^w \,\|\, \mathbf{d}_2^r] \in \mathbb{R}^{B\times 256\times \frac{H}{2}\times \frac{W}{2}}$

// Multi-Scale HoVer-Net Nuclear Distance Prediction Head:

$\mathbf{d}_h^{(1)} \leftarrow \mathbf{D}_1[:, :64], \quad \mathbf{d}_v^{(1)} \leftarrow \mathbf{D}_1[:, 64:128]$ ▷ Retrieve Stage 1 horizontal and vertical detail bands
$\mathbf{F}_{\text{hv}} \leftarrow \left[\mathbf{d}_1^w \,\|\, \mathbf{d}_1^r \,\|\, \text{BilinearInterpolate}(\mathbf{d}_h^{(1)}, (H, W)) \,\|\, \text{BilinearInterpolate}(\mathbf{d}_v^{(1)}, (H, W))\right] \in \mathbb{R}^{B\times 256\times H\times W}$
$\hat{\mathbf{H}}, \hat{\mathbf{V}} \leftarrow \tanh\left(\text{Conv}_{1\times 1}\left(\text{ReLU}(\text{Conv}_{3\times 3}(\mathbf{F}_{\text{hv}}))\right)\right)$ ▷ Distance-to-centroid maps in $[-1,1]$

return $\{\mathbf{z}_{\text{fused}}, \hat{\mathbf{H}}, \hat{\mathbf{V}}, \mathcal{L}_{\text{feat}}, \mathbf{z}_{\text{aux3}}, \mathbf{z}_{\text{aux2}}\}$

Algorithm 2 Polyphase Learnable Lifting Wavelet Forward Decomposition (LLWT)

Require: Feature tensor $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$

Ensure: Low-frequency coarse approximation $c_l \in \mathbb{R}^{B\times C\times \frac{H}{2}\times \frac{W}{2}}$, High-frequency detail bypass $\mathbf{D} \in \mathbb{R}^{B\times 3C\times \frac{H}{2}\times \frac{W}{2}}$, Downsampled pooled output tensor $\mathbf{X}_{\text{pooled}} \in \mathbb{R}^{B\times 2C\times \frac{H}{2}\times \frac{W}{2}}$

// Step 1: Polyphase Sub-Grid Splitting

Extract four non-overlapping spatial sub-grids of shape $\mathbb{R}^{B\times C\times \frac{H}{2}\times \frac{W}{2}}$:

$x_{00}(b,c,i,j) \leftarrow \mathbf{X}(b,c,2i,2j)$ ▷ Even rows, even columns (reference coset)

$x_{01}(b,c,i,j) \leftarrow \mathbf{X}(b,c,2i,2j+1)$ ▷ Even rows, odd columns (horizontal phase)

$x_{10}(b,c,i,j) \leftarrow \mathbf{X}(b,c,2i+1,2j)$ ▷ Odd rows, even columns (vertical phase)

$x_{11}(b,c,i,j) \leftarrow \mathbf{X}(b,c,2i+1,2j+1)$ ▷ Odd rows, odd columns (diagonal phase)

// Step 2: StainAdapt Feature Modulation

$\mathbf{y}_{\text{gap}} \leftarrow \text{GAP}(x_{00}) \in \mathbb{R}^{B\times C\times 1\times 1}$ ▷ Global average pooling descriptor

for $k \in \{h, v, d\}$ **do**

$\mathbf{s}_k \leftarrow \sigma\Big(\mathbf{W}_{2,k} \cdot \text{ReLU}\left(\mathbf{W}_{1,k}\mathbf{y}_{\text{gap}} + \mathbf{b}_{1,k}\right) + \mathbf{b}_{2,k}\Big)$ ▷ Two-layer MLP with reduction ratio $r = 4$

$S_k(x_{00}, f) \leftarrow f \odot \mathbf{s}_k$ ▷ Channel-wise broadcast multiplication

end for

// Step 3: Directional Depthwise Predict Step

$d_h \leftarrow x_{01} - S_h\big(x_{00}, \mathcal{P}_h(x_{00})\big)$ ▷ Horizontal high-frequency detail sub-band

$d_v \leftarrow x_{10} - S_v\big(x_{00}, \mathcal{P}_v(x_{00})\big)$ ▷ Vertical high-frequency detail sub-band

$d_d \leftarrow x_{11} - S_d\big(x_{00}, \mathcal{P}_d(x_{00})\big)$ ▷ Diagonal high-frequency detail sub-band

// Step 4: Directional Depthwise Update Step

$c_l \leftarrow x_{00} + k_h\mathcal{U}_h(d_h) + k_v\mathcal{U}_v(d_v) + k_d\mathcal{U}_d(d_d)$ ▷ Low-frequency coarse approximation

// Step 5: Sub-Band Aggregation and Bypass Construction

$\mathbf{D} \leftarrow [d_h \,\|\, d_v \,\|\, d_d] \in \mathbb{R}^{B\times 3C\times \frac{H}{2}\times \frac{W}{2}}$ ▷ Preserved detail bypass routed directly to decoder and HoVer head

$\mathbf{X}_{\text{sub}} \leftarrow [c_l \,\|\, \mathbf{D}] \in \mathbb{R}^{B\times 4C\times \frac{H}{2}\times \frac{W}{2}}$ ▷ Concatenate approximation and detail sub-bands

$\mathbf{X}_{\text{pooled}} \leftarrow \text{ReLU}\Big(\text{BN}\left(\text{Conv}_{1\times 1}^{4C\to 2C}(\mathbf{X}_{\text{sub}})\right)\Big)$ ▷ Channel projection to next stage width

return $\{c_l, \mathbf{D}, \mathbf{X}_{\text{pooled}}\}$

Algorithm 3 Inverse Learnable Lifting Wavelet Decoding (ILLWT)

Require: Incoming decoder feature tensor $\mathbf{x}_{\text{dec}} \in \mathbb{R}^{B \times C_{\text{in}} \times H \times W}$,
Detail bypass $\mathbf{D} = [d_h \parallel d_v \parallel d_d] \in \mathbb{R}^{B \times 3C_{\text{skip}} \times H \times W}$,
Encoder skip connection tensor $\mathbf{s}_{\text{skip}} \in \mathbb{R}^{B \times C_{\text{skip}} \times 2H \times 2W}$
Ensure: Reconstructed and merged feature tensor $\mathbf{y}_{\text{dec}} \in \mathbb{R}^{B \times C_{\text{out}} \times 2H \times 2W}$

// Step 1: Channel Width Projection
$c_l \leftarrow \text{Conv}_{1\times1}^{C_{\text{in}} \to C_{\text{skip}}}(\mathbf{x}_{\text{dec}}) \in \mathbb{R}^{B \times C_{\text{skip}} \times H \times W}$ ▷ Project decoder channels to match skip dimension

// Step 2: Inverse Update and Predict Equations
$x_{00} \leftarrow c_l - \left(l_h \mathcal{U}'_h(d_h) + l_v \mathcal{U}'_v(d_v) + l_d \mathcal{U}'_d(d_d)\right)$ ▷ Recover reference polyphase coset exactly
$x_{01} \leftarrow d_h + \mathcal{P}'_h(x_{00})$ ▷ Recover horizontal phase
$x_{10} \leftarrow d_v + \mathcal{P}'_v(x_{00})$ ▷ Recover vertical phase
$x_{11} \leftarrow d_d + \mathcal{P}'_d(x_{00})$ ▷ Recover diagonal phase

// Step 3: Cartesian Spatial Lattice Interlacing
Allocate empty tensor $\mathbf{X}_{\text{recon}} \in \mathbb{R}^{B \times C_{\text{skip}} \times 2H \times 2W}$:
$\mathbf{X}_{\text{recon}}(:, :, 2i, 2j) \leftarrow x_{00}(:, :, i, j)$
$\mathbf{X}_{\text{recon}}(:, :, 2i, 2j+1) \leftarrow x_{01}(:, :, i, j)$
$\mathbf{X}_{\text{recon}}(:, :, 2i+1, 2j) \leftarrow x_{10}(:, :, i, j)$
$\mathbf{X}_{\text{recon}}(:, :, 2i+1, 2j+1) \leftarrow x_{11}(:, :, i, j)$

// Step 4: Skip Merging and Residual Processing
$\tilde{\mathbf{s}} \leftarrow \mathbf{s}_{\text{skip}} + \mathbf{X}_{\text{recon}}$ ▷ Additive injection of high-frequency reconstructed edges
$\mathbf{u}_{\text{dec}} \leftarrow \text{BilinearInterpolate}(\mathbf{x}_{\text{dec}}, \text{ size} = (2H, 2W))$ ▷ Upsample incoming decoder features
$\mathbf{F}_{\text{cat}} \leftarrow [\mathbf{u}_{\text{dec}} \parallel \tilde{\mathbf{s}}] \in \mathbb{R}^{B \times (C_{\text{in}} + C_{\text{skip}}) \times 2H \times 2W}$ ▷ Concatenate semantic context and edge features
$\mathbf{F}_{\text{merged}} \leftarrow \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}(\mathbf{F}_{\text{cat}})\right)\right)$ ▷ Merge feature channels
$\mathbf{y}_{\text{dec}} \leftarrow \text{ResBlocks}(\mathbf{F}_{\text{merged}})$ ▷ Refine via Pre-activation Residual Blocks
return $\mathbf{y}_{\text{dec}}$

3. DOWNSAMPLING, BOTTLENECKS, AND CROSS-ATTENTION

This section formalizes the downsampling, bottleneck, and feature synchronization operators that maintain representation parity between the two streams.

Algorithm 4 Generalized Mean (GeM) Pooling Downsample

Require: Feature tensor $\mathbf{X} \in \mathbb{R}^{B \times C \times H \times W}$,
Unconstrained learnable exponent $p \in [1, \infty)$ (initialized to 3.0), Stability constant $\epsilon = 10^{-6}$
Ensure: Downsampled feature tensor $\mathbf{X}_{\text{out}} \in \mathbb{R}^{B \times 2C \times \frac{H}{2} \times \frac{W}{2}}$

$\mathbf{X}_{\text{clamped}} \leftarrow \text{clamp}(\mathbf{X}, \text{min} = \epsilon)$ ▷ Ensure strictly positive base values
$\mathbf{X}_{\text{pow}} \leftarrow \mathbf{X}_{\text{clamped}}^{p}$ ▷ Element-wise power evaluation
$\mathbf{X}_{\text{avg}} \leftarrow \text{AvgPool2d}(\mathbf{X}_{\text{pow}}, \text{kernel_size} = 2, \text{stride} = 2)$ ▷ Spatial window summation $\frac{1}{|\Omega|} \sum_{i \in \Omega} x_i^p$
$\mathbf{X}_{\text{gem}} \leftarrow \text{clamp}(\mathbf{X}_{\text{avg}}, \text{min} = \epsilon)^{1/p}$ ▷ Root-p inverse exponentiation
$\mathbf{X}_{\text{out}} \leftarrow \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}^{C \to 2C}(\mathbf{X}_{\text{gem}})\right)\right)$ ▷ Channel projection to next stage width
return $\mathbf{X}_{\text{out}}$

Algorithm 5 Atrous Spatial Pyramid Pooling (ASPP) Bottleneck with Dilation Rates $\{1, 4, 7\}$

Require: Bottleneck feature tensor $\mathbf{X} \in \mathbb{R}^{B \times C \times H \times W}$ $(C = 1024)$
Ensure: Multi-scale contextual tensor $\mathbf{X}_{\text{out}} \in \mathbb{R}^{B \times C \times H \times W}$

// Branch 0: 1×1 Convolutional Branch
$\mathbf{b}_0 \leftarrow \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}^{C \to C/4}(\mathbf{X})\right)\right) \in \mathbb{R}^{B \times 256 \times H \times W}$

// Branches 1, 2, 3: Depthwise-Separable Dilated Convolutions
for $r \in \{1, 4, 7\}$ **do**
    $\mathbf{t}_{\text{dw}} \leftarrow \text{DWConv}_{3\times3,\ \text{dilation}=r,\ \text{padding}=r}(\mathbf{X})$ ▷ Spatial filtering per channel (groups $= C$)
    $\mathbf{b}_r \leftarrow \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}^{C \to C/4}(\text{ReLU}(\text{BN}(\mathbf{t}_{\text{dw}})))\right)\right)$ ▷ Pointwise projection to 256 channels
end for

// Branch 4: Image Global Average Pooling Branch
$\mathbf{y}_{\text{gap}} \leftarrow \text{AdaptiveAvgPool2D}(\mathbf{X}, \text{output_size} = (1, 1)) \in \mathbb{R}^{B \times C \times 1 \times 1}$
$\mathbf{y}_{\text{proj}} \leftarrow \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}^{C \to C/4}(\mathbf{y}_{\text{gap}})\right)\right) \in \mathbb{R}^{B \times 256 \times 1 \times 1}$
$\mathbf{b}_{\text{pool}} \leftarrow \text{BilinearInterpolate}(\mathbf{y}_{\text{proj}}, \text{size} = (H, W)) \in \mathbb{R}^{B \times 256 \times H \times W}$

// Feature Concatenation, Projection, and Spatial Dropout
$\mathbf{B}_{\text{cat}} \leftarrow [\mathbf{b}_0 \parallel \mathbf{b}_1 \parallel \mathbf{b}_4 \parallel \mathbf{b}_7 \parallel \mathbf{b}_{\text{pool}}] \in \mathbb{R}^{B \times 1280 \times H \times W}$
$\mathbf{X}_{\text{proj}} \leftarrow \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}^{1280 \to 1024}(\mathbf{B}_{\text{cat}})\right)\right)$
$\mathbf{X}_{\text{out}} \leftarrow \text{Dropout2D}(\mathbf{X}_{\text{proj}}, p = 0.1)$ ▷ Spatial dropout regularizing co-adaptation
return $\mathbf{X}_{\text{out}}$

Algorithm 6 Multi-Scale Bi-Directional Channel Cross-Attention (BiCA)

Require: Input feature streams $\mathbf{x}_w, \mathbf{x}_r \in \mathbb{R}^{B\times C\times H\times W}$
Ensure: Cross-calibrated output streams $\hat{\mathbf{x}}_w, \hat{\mathbf{x}}_r \in \mathbb{R}^{B\times C\times H\times W}$

function CROSSATTENTIONSTREAM($\mathbf{x}_{\text{query}}, \mathbf{x}_{\text{keyval}}$)
  $N \leftarrow H \cdot W$ ▷ Spatial token count
  $\mathbf{Q} \leftarrow \text{Reshape}\Big(\text{BN}\big(\text{Conv}_{1\times1}(\mathbf{x}_{\text{query}})\big),\ (B, C, N)\Big)$ ▷ Query projection
  $\mathbf{K} \leftarrow \text{Reshape}\Big(\text{BN}\big(\text{Conv}_{1\times1}(\mathbf{x}_{\text{keyval}})\big),\ (B, C, N)\Big)$ ▷ Key projection
  $\mathbf{V} \leftarrow \text{Reshape}\Big(\text{Conv}_{1\times1}(\mathbf{x}_{\text{keyval}}),\ (B, C, N)\Big)$ ▷ Value projection
  $\mathbf{A} \leftarrow \frac{1}{\sqrt{N}}\big(\mathbf{Q}\mathbf{K}^\top\big) \in \mathbb{R}^{B\times C\times C}$ ▷ Channel cross-affinity matrix scaled by $1/\sqrt{N}$
  $\tilde{\mathbf{A}} \leftarrow \text{Softmax}(\mathbf{A},\ \text{dim} = -1) \in \mathbb{R}^{B\times C\times C}$ ▷ Row-wise normalized attention weights
  $\mathbf{O} \leftarrow \tilde{\mathbf{A}}\mathbf{V} \in \mathbb{R}^{B\times C\times N}$ ▷ Attended channel aggregation
  $\mathbf{O}_{\text{spatial}} \leftarrow \text{Reshape}(\mathbf{O},\ (B, C, H, W))$
  $\mathbf{W}_{\text{out}} \leftarrow \text{BN}_{(\gamma=0,\beta=0)}\big(\text{Conv}_{1\times1}(\mathbf{O}_{\text{spatial}})\big)$ ▷ Zero-initialized BatchNorm projection
  return $\mathbf{x}_{\text{query}} + \mathbf{W}_{\text{out}}$ ▷ Exact identity operator at initialization step 0
end function

$\hat{\mathbf{x}}_w \leftarrow$ CROSSATTENTIONSTREAM($\mathbf{x}_w, \mathbf{x}_r$) ▷ Query: Wavelet stream, Key/Value: ResNet stream
$\hat{\mathbf{x}}_r \leftarrow$ CROSSATTENTIONSTREAM($\mathbf{x}_r, \mathbf{x}_w$) ▷ Query: ResNet stream, Key/Value: Wavelet stream
return $\{\hat{\mathbf{x}}_w,\ \hat{\mathbf{x}}_r\}$

Algorithm 7 Skip-Connection Feature Alignment Loss ($\mathcal{L}_{\text{feat}}$)

Require: ResUNet decoder transposed-convolution tensors $\{\mathbf{u}_l^r\}_{l=1}^4$,
ResUNet encoder skip connection tensors $\{\mathbf{s}_l^r\}_{l=1}^4$, Numerical epsilon $\epsilon = 10^{-6}$
Ensure: Normalized scalar alignment loss $\mathcal{L}_{\text{feat}} \in \mathbb{R}$

$\mathcal{L}_{\text{feat}} \leftarrow 0.0$
for $l \leftarrow 1$ **to** 4 **do**
  Compute spatial channel-wise L_2 norm vectors:
  $\mathbf{n}_{\mathbf{u}}(b, 1, i, j) \leftarrow \sqrt{\sum_{c=1}^{C_l} \mathbf{u}_l^r(b, c, i, j)^2} + \epsilon, \quad \forall(b, i, j)$
  $\mathbf{n}_{\mathbf{s}}(b, 1, i, j) \leftarrow \sqrt{\sum_{c=1}^{C_l} \mathbf{s}_l^r(b, c, i, j)^2} + \epsilon, \quad \forall(b, i, j)$
  $\hat{\mathbf{u}}_l^r \leftarrow \mathbf{u}_l^r \,/\, \mathbf{n}_{\mathbf{u}}, \quad \hat{\mathbf{s}}_l^r \leftarrow \mathbf{s}_l^r \,/\, \mathbf{n}_{\mathbf{s}}$ ▷ Normalize channel vectors to unit sphere
  $\mathcal{E}_l \leftarrow \frac{1}{B\cdot H_l\cdot W_l} \sum_{b,c,i,j} \Big(\hat{\mathbf{u}}_l^r(b, c, i, j) - \hat{\mathbf{s}}_l^r(b, c, i, j)\Big)^2$ ▷ Mean squared error across stage l
  $\mathcal{L}_{\text{feat}} \mathrel{+}= \mathcal{E}_l$ ▷ Accumulate across all four hierarchy levels
end for
return $\mathcal{L}_{\text{feat}}$ ▷ Minimizes representation divergence across skip bridges

4. FUZZY FUSION AND SUPERVISION OBJECTIVES

This section formalizes the Sugeno fuzzy integral ensembling mechanism and the multi-task composite loss evaluation.

Algorithm 8 Learnable Sugeno Fuzzy Decision Fusion

Require: Wavelet stream probability map $p_w \in [0,1]^{B\times 1\times H\times W}$, ResNet stream probability map $p_r \in [0,1]^{B\times 1\times H\times W}$,
Unconstrained learnable logit parameters $\theta_1, \theta_2 \in \mathbb{R}$ (initialized to 0.0), Numerical constant $\epsilon = 10^{-6}$
Ensure: Ensembled probability map $p_{\text{fused}} \in [0,1]^{B\times 1\times H\times W}$, Fused logit tensor $\mathbf{z}_{\text{fused}} \in \mathbb{R}^{B\times 1\times H\times W}$

$g_1 \leftarrow \sigma(\theta_1) = \frac{1}{1+e^{-\theta_1}}, \quad g_2 \leftarrow \sigma(\theta_2) = \frac{1}{1+e^{-\theta_2}}$ ▷ Evaluate singleton fuzzy densities in $(0,1)$
$\mathbf{M} \leftarrow \mathbb{I}(p_w \geq p_r)$ ▷ Spatial indicator mask where Wavelet branch dominates
$h_{(1)} \leftarrow \max(p_w, p_r), \quad h_{(2)} \leftarrow \min(p_w, p_r)$ ▷ Rank-ordered branch evidence
$g_{\max} \leftarrow \mathbf{M} \cdot g_1 + (1-\mathbf{M}) \cdot g_2$ ▷ Select singleton density of dominant branch
$T_1 \leftarrow \min(h_{(1)}, g_{\max})$ ▷ First Sugeno fuzzy term: $h_{(1)} \wedge g(F_1)$
$T_2 \leftarrow h_{(2)}$ ▷ Second Sugeno fuzzy term: $h_{(2)} \wedge g(X) = \min(h_{(2)}, 1) = h_{(2)}$
$p_{\text{fused}} \leftarrow \max(T_1, T_2)$ ▷ Closed-form analytical Sugeno fuzzy integral
$p_{\text{clamped}} \leftarrow \text{clamp}(p_{\text{fused}}, \epsilon, 1.0-\epsilon)$ ▷ Prevent numerical divergence in logit transformation
$\mathbf{z}_{\text{fused}} \leftarrow \ln(p_{\text{clamped}}) - \ln(1.0 - p_{\text{clamped}})$ ▷ Inverse sigmoid reconstruction for BCEWithLogitsLoss
return $\{p_{\text{fused}}, \mathbf{z}_{\text{fused}}\}$

Algorithm 9 Multi-Task Composite Loss Evaluation

Require: Primary fused logits $\mathbf{z}_{\text{fused}}$, Auxiliary logits $\mathbf{z}_{\text{aux3}}, \mathbf{z}_{\text{aux2}}$, Distance maps $\hat{\mathbf{H}}, \hat{\mathbf{V}}$, Skip alignment loss $\mathcal{L}_{\text{feat}}$,
Ground truth binary mask $\mathbf{y} \in \{0,1\}^{B\times 1\times H\times W}$, Ground truth distance maps $\mathbf{H}^*, \mathbf{V}^* \in [-1,1]^{B\times 1\times H\times W}$
Ensure: Total composite loss scalar $\mathcal{L}_{\text{total}} \in \mathbb{R}$

function SEGLOSS($\mathbf{z}, \mathbf{y}$)
  $\mathbf{p} \leftarrow \sigma(\mathbf{z})$
  $TP \leftarrow \sum \mathbf{p}\mathbf{y}, \quad FN \leftarrow \sum (1-\mathbf{p})\mathbf{y}, \quad FP \leftarrow \sum \mathbf{p}(1-\mathbf{y})$
  $TI \leftarrow \frac{TP+1.0}{TP+0.7FN+0.3FP+1.0}$ ▷ Tversky Index with $\alpha = 0.7$ and $\beta = 0.3$
  $\mathcal{L}_{\text{FT}} \leftarrow (1.0 - TI)^{0.75}$ ▷ Focal exponent $\gamma = 0.75$
  $w_{\text{pos}} \leftarrow \min\left(\frac{\sum(1-\mathbf{y})}{\sum \mathbf{y}+10^{-5}}, 15.0\right)$ ▷ Dynamic background-to-foreground class weighting
  $\mathcal{L}_{\text{BCE}} \leftarrow \text{BCEWithLogits}(\mathbf{z}, \mathbf{y}, \text{pos_weight} = w_{\text{pos}})$
  return $0.6 \cdot \mathcal{L}_{\text{FT}} + 0.4 \cdot \mathcal{L}_{\text{BCE}}$
end function

$\mathcal{L}_{\text{seg}} \leftarrow \text{SEGLOSS}(\mathbf{z}_{\text{fused}}, \mathbf{y}) + 0.4\text{SEGLOSS}(\mathbf{z}_{\text{aux3}}, \mathbf{y}) + 0.2\text{SEGLOSS}(\mathbf{z}_{\text{aux2}}, \mathbf{y})$ ▷ Deep supervision

// HoVer-Net Nuclear Distance Regression Loss
$\mathcal{L}_{\text{mse}} \leftarrow \text{MSE}(\hat{\mathbf{H}}, \mathbf{H}^*) + \text{MSE}(\hat{\mathbf{V}}, \mathbf{V}^*)$
$\mathcal{L}_{\text{msge}} \leftarrow \text{MSGE}(\hat{\mathbf{H}}, \mathbf{H}^*, \mathbf{y}) + \text{MSGE}(\hat{\mathbf{V}}, \mathbf{V}^*, \mathbf{y})$ ▷ Foreground-restricted Sobel gradient error
$\mathcal{L}_{\text{hv}} \leftarrow 1.0 \cdot \mathcal{L}_{\text{mse}} + 1.0 \cdot \mathcal{L}_{\text{msge}}$

// Differentiable Boundary Consistency Loss
function BOUNDARYLOSS($\mathbf{z}, \mathbf{y}$)
  $|\nabla_S \mathbf{P}| \leftarrow \sqrt{(\sigma(\mathbf{z}) * \mathbf{K}_x)^2 + (\sigma(\mathbf{z}) * \mathbf{K}_y)^2 + 10^{-6}}$ ▷ Predicted edge magnitude
  $|\nabla_S \mathbf{Y}| \leftarrow \sqrt{(\mathbf{y} * \mathbf{K}_x)^2 + (\mathbf{y} * \mathbf{K}_y)^2 + 10^{-6}}$ ▷ Ground truth edge magnitude
  return $\frac{1}{N}\sum \left| |\nabla_S \mathbf{P}| - |\nabla_S \mathbf{Y}| \right|$ ▷ Mean absolute gradient discrepancy
end function
$\mathcal{L}_{\text{bnd}} \leftarrow \text{BOUNDARYLOSS}(\mathbf{z}_{\text{fused}}, \mathbf{y}) + 0.4\text{BOUNDARYLOSS}(\mathbf{z}_{\text{aux3}}, \mathbf{y}) + 0.2\text{BOUNDARYLOSS}(\mathbf{z}_{\text{aux2}}, \mathbf{y})$

$\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{seg}} + 0.1 \cdot \mathcal{L}_{\text{feat}} + 1.0 \cdot \mathcal{L}_{\text{hv}} + 0.15 \cdot \mathcal{L}_{\text{bnd}}$ ▷ Composite objective
return $\mathcal{L}_{\text{total}}$

5. OPTIMIZATION DYNAMICS AND TILE INFERENCE

This section details the end-to-end training loop and the artifact-free sliding-window inference pipeline.

Algorithm 10 End-to-End Training Loop with AdamW, EMA, and OneCycleLR

Require: Dataset $\mathcal{D}$, Total Epochs $E = 500$, Initial Learning Rate $\eta = 3 \times 10^{-4}$, Weight Decay $\lambda = 10^{-3}$, Batch Size $B = 4$
Ensure: Trained model parameters $\boldsymbol{\theta}_{\text{EMA}}$

Initialize network weights $\boldsymbol{\theta}$; Initialize shadow weights $\boldsymbol{\theta}_{\text{EMA}} \leftarrow \text{deepcopy}(\boldsymbol{\theta})$
Initialize AdamW optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $\lambda = 10^{-3}$)
Initialize OneCycleLR scheduler (max_lr $= \eta$, epochs $= E$, pct_start $= 0.3$, cosine annealing)
Initialize dynamic gradient scaler scaler $\leftarrow \text{GradScaler}()$; $t_{\text{step}} \leftarrow 0$
for epoch $\leftarrow 1$ **to** E **do**
  for batch $(\mathbf{X}, \mathbf{y}, \mathbf{H}^*, \mathbf{V}^*)$ in $\mathcal{D}$ **do**
    With Automatic Mixed Precision (AMP):
      Forward pass: $\mathcal{L}_{\text{total}} \leftarrow \text{EvaluateLoss}(\mathcal{M}_{\boldsymbol{\theta}}(\mathbf{X}),\ \mathbf{y},\ \mathbf{H}^*,\ \mathbf{V}^*)$
    Scaled backpropagation: scaler. scale($\mathcal{L}_{\text{total}}$).backward()
    scaler. unscale(optimizer)
    Gradient clipping: $\mathbf{g} \leftarrow \text{clip_grad_norm}(\boldsymbol{\theta},\ \text{max_norm} = 1.0)$
    scaler. step(optimizer); scaler. update()
    optimizer.zero_grad(); scheduler.step()

    // Exponential Moving Average Shadow Weight Update:
    t_{step} += 1
    $\alpha_t \leftarrow \min\left(0.999,\ \frac{1+t_{\text{step}}}{10+t_{\text{step}}}\right)$ ▷ Warmup schedule transitioning to asymptotic decay $\alpha = 0.999$
    $\boldsymbol{\theta}_{\text{EMA}} \leftarrow \alpha_t \boldsymbol{\theta}_{\text{EMA}} + (1 - \alpha_t)\boldsymbol{\theta}$
  end for
end for
return $\boldsymbol{\theta}_{\text{EMA}}$

Algorithm 11 Seamless Full-Image Sliding-Window Inference via Raised-Cosine Blending

Require: Full tissue image $\mathbf{I}_{\text{full}} \in \mathbb{R}^{3 \times H_{\text{full}} \times W_{\text{full}}}$, Patch dimension $N = 256$, Stride $S = 192$ (giving $\approx 25\%$ spatial overlap)
Ensure: Full probability map $\hat{\mathbf{P}}_{\text{full}} \in [0,1]^{1 \times H_{\text{full}} \times W_{\text{full}}}$, Full distance maps $\hat{\mathbf{H}}_{\text{full}}, \hat{\mathbf{V}}_{\text{full}} \in [-1,1]^{1 \times H_{\text{full}} \times W_{\text{full}}}$

Construct 1D raised-cosine Hanning window with lower bound 0.1:
$w(n) \leftarrow \max\left(0.5\big(1 - \cos(\frac{2\pi n}{N-1})\big),\ 0.1\right), \quad \forall n \in \{0, \ldots, N-1\}$
Form 2D outer-product blending kernel: $\mathbf{W}(i,j) \leftarrow w(i) \cdot w(j) \in \mathbb{R}^{N \times N}$
Initialize accumulator tensors $\mathbf{A}_{\text{prob}} \leftarrow \mathbf{0}_{1 \times H_{\text{full}} \times W_{\text{full}}}$, $\mathbf{A}_{\text{hv}} \leftarrow \mathbf{0}_{2 \times H_{\text{full}} \times W_{\text{full}}}$, $\boldsymbol{\Omega} \leftarrow \mathbf{0}_{1 \times H_{\text{full}} \times W_{\text{full}}}$
for each coordinate (y, x) generated by sliding stride S **do**
  Extract native patch: $\mathbf{X}_{\text{patch}} \leftarrow \mathbf{I}_{\text{full}}[:, y : y+N, x : x+N]$
  $\{\mathbf{z}_{\text{fused}}, \hat{\mathbf{H}}, \hat{\mathbf{V}}\} \leftarrow \mathcal{M}_{\boldsymbol{\theta}_{\text{EMA}}}(\mathbf{X}_{\text{patch}})$
  $\mathbf{P}_{\text{patch}} \leftarrow \sigma(\mathbf{z}_{\text{fused}})$
  $\mathbf{A}_{\text{prob}}[:, y : y+N, x : x+N]$ += $\mathbf{P}_{\text{patch}} \odot \mathbf{W}$
  $\mathbf{A}_{\text{hv}}[:, y : y+N, x : x+N]$ += $[\hat{\mathbf{H}} \,\|\, \hat{\mathbf{V}}] \odot \mathbf{W}$
  $\boldsymbol{\Omega}[:, y : y+N, x : x+N]$ += $\mathbf{W}$
end for
$\hat{\mathbf{P}}_{\text{full}} \leftarrow \mathbf{A}_{\text{prob}} \,/\, \max(\boldsymbol{\Omega}, 10^{-6})$ ▷ Normalized blended probability map
$\hat{\mathbf{H}}_{\text{full}}, \hat{\mathbf{V}}_{\text{full}} \leftarrow \mathbf{A}_{\text{hv}} \,/\, \max(\boldsymbol{\Omega}, 10^{-6})$ ▷ Normalized blended distance maps
return $\{\hat{\mathbf{P}}_{\text{full}},\ \hat{\mathbf{H}}_{\text{full}},\ \hat{\mathbf{V}}_{\text{full}}\}$

6. INSTANCE DECODING AND BENCHMARK EVALUATION

This section details the marker-controlled watershed instance segmentation decoding pipeline, hyperparameter search, and standard benchmark evaluation metrics.

Algorithm 12 Marker-Controlled Distance-Guided Watershed Instance Decoding

Require: Full probability $\hat{\mathbf{P}} \in [0,1]$, Distance maps $\hat{\mathbf{H}}, \hat{\mathbf{V}} \in [-1,1]$, Thresholds $\tau_{\text{thresh}} = 0.5, \tau_{\text{seed}}, r_{\text{erode}}$, Minimum instance area $A_{\min} = 10$
Ensure: Multi-instance labeled segmentation mask $\mathbf{Y}_{\text{inst}} \in \mathbb{N}_0^{H \times W}$

// Step 1: Topological Gradient Energy Landscape
$\mathbf{G}_h \leftarrow \sqrt{(\hat{\mathbf{H}} * \mathbf{K}_x)^2 + (\hat{\mathbf{H}} * \mathbf{K}_y)^2}, \quad \mathbf{G}_v \leftarrow \sqrt{(\hat{\mathbf{V}} * \mathbf{K}_x)^2 + (\hat{\mathbf{V}} * \mathbf{K}_y)^2}$ ▷ Sobel spatial gradients
$\mathbf{G} \leftarrow \max(\mathbf{G}_h, \mathbf{G}_v)$ ▷ Combined gradient magnitude
$\tilde{\mathbf{G}} \leftarrow \frac{\mathbf{G} - \min(\mathbf{G})}{\max(\mathbf{G}) - \min(\mathbf{G}) + 10^{-8}} \in [0,1]$ ▷ Normalized gradient surface (energy basins at centroids)

// Step 2: Seed Extraction and Morphological Erosion
$\mathbf{M}_{\text{bin}} \leftarrow (\hat{\mathbf{P}} \geq \tau_{\text{thresh}})$ ▷ Primary foreground mask
$\mathbf{S}_{\text{seed}} \leftarrow (\tilde{\mathbf{G}} < \tau_{\text{seed}}) \wedge \mathbf{M}_{\text{bin}}$ ▷ Extract low-gradient nuclear centroids
if $r_{\text{erode}} > 0$ **then**
  $\mathcal{B} \leftarrow \text{DiskStructuringElement}(r_{\text{erode}})$
  $\mathbf{S}_{\text{seed}} \leftarrow \mathbf{S}_{\text{seed}} \ominus \mathcal{B}$ ▷ Morphological erosion cleanly detaching abutting seeds
end if
$\mathbf{M}_{\text{markers}} \leftarrow \text{LabelConnectedComponents}(\mathbf{S}_{\text{seed}})$ ▷ Assign unique integer labels to disjoint seeds

// Step 3: Marker-Controlled Watershed Flooding
if $\max(\mathbf{M}_{\text{markers}}) == 0$ **then**
  $\mathbf{Y}_{\text{inst}} \leftarrow \text{ConnectedComponents}(\mathbf{M}_{\text{bin}})$ ▷ Fallback if no seeds survive
else
  $\mathbf{Y}_{\text{inst}} \leftarrow \text{Watershed}(\tilde{\mathbf{G}}, \mathbf{M}_{\text{markers}}, \text{mask} = \mathbf{M}_{\text{bin}})$ ▷ Flood from markers across landscape $\tilde{\mathbf{G}}$
end if

// Step 4: Spurious Small Instance Pruning
for each unique label $k \in \mathbf{Y}_{\text{inst}} \setminus \{0\}$ **do**
  if $\sum \mathbb{I}(\mathbf{Y}_{\text{inst}} == k) < A_{\min}$ **then**
    $\mathbf{Y}_{\text{inst}}[\mathbf{Y}_{\text{inst}} == k] \leftarrow 0$ ▷ Prune debris smaller than 10 pixels
  end if
end for
return $\mathbf{Y}_{\text{inst}}$

Algorithm 13 Validation Grid Search for Watershed Hyperparameters

Require: Validation dataset $\mathcal{D}_{\text{val}} = \{(\mathbf{I}_k, \mathbf{Y}_k^*)\}_{k=1}^{V}$, Trained model $\mathcal{M}$
Ensure: Optimal watershed hyperparameter configuration $(\tau_{\text{seed}}^*, r_{\text{erode}}^*)$

$\mathcal{T} \leftarrow \{0.3, 0.4, 0.5, 0.6\}$ ▷ Candidate seed threshold values
$\mathcal{R} \leftarrow \{0, 1, 2, 3\}$ ▷ Candidate disk erosion radii
BestAJI $\leftarrow -1.0, \quad (\tau_{\text{seed}}^*, r_{\text{erode}}^*) \leftarrow (0.4, 1)$
Pre-compute inference maps $\{\hat{\mathbf{P}}_k, \hat{\mathbf{H}}_k, \hat{\mathbf{V}}_k\}$ for all validation images $k \in \{1, \ldots, V\}$
for $\tau \in \mathcal{T}$ **do**
  for $r \in \mathcal{R}$ **do**
    $\text{AJI}_{\text{sum}} \leftarrow 0.0$
    for $k \leftarrow 1$ **to** V **do**
      $\mathbf{Y}_{\text{pred}} \leftarrow \text{WatershedDecode}(\hat{\mathbf{P}}_k, \hat{\mathbf{H}}_k, \hat{\mathbf{V}}_k, \tau_{\text{seed}} = \tau, r_{\text{erode}} = r)$
      $\text{AJI}_{\text{sum}} \mathrel{+}= \text{EvaluateAJI}(\mathbf{Y}_{\text{pred}}, \mathbf{Y}_k^*)$
    end for
    $\text{MeanAJI} \leftarrow \text{AJI}_{\text{sum}} / V$
    if MeanAJI $>$ BestAJI **then**
      BestAJI $\leftarrow$ MeanAJI
      $(\tau_{\text{seed}}^*, r_{\text{erode}}^*) \leftarrow (\tau, r)$
    end if
  end for
end for
return $(\tau_{\text{seed}}^*, r_{\text{erode}}^*)$

Algorithm 14 Aggregated Jaccard Index (AJI) Evaluation

Require: Predicted instance segmentation mask $\mathbf{Y}_{\text{pred}} \in \mathbb{N}_0^{H \times W}$, Ground truth instance mask $\mathbf{Y}_{\text{gt}} \in \mathbb{N}_0^{H \times W}$
Ensure: Scalar AJI metric score $\in [0, 1]$

Extract ground truth instance labels: $\mathcal{G} \leftarrow \text{unique}(\mathbf{Y}_{\text{gt}}) \setminus \{0\} = \{G_1, \ldots, G_K\}$
Extract predicted instance labels: $\mathcal{P} \leftarrow \text{unique}(\mathbf{Y}_{\text{pred}}) \setminus \{0\} = \{P_1, \ldots, P_M\}$
if $|\mathcal{G}| == 0 \land |\mathcal{P}| == 0$ **then return** 1.0 ▷ Empty image correctly identified
end if
if $|\mathcal{G}| == 0 \lor |\mathcal{P}| == 0$ **then return** 0.0 ▷ Extreme under- or over-segmentation penalty
end if
IntersectionSum $\leftarrow 0$, UnionSum $\leftarrow 0$
$\mathcal{U}_{\text{pred}} \leftarrow \text{set}(\mathcal{P})$ ▷ Track unmatched predictions
for $i \leftarrow 1$ **to** K **do**
    $\mathbf{M}_g \leftarrow (\mathbf{Y}_{\text{gt}} == G_i)$ ▷ Binary mask of ground truth nucleus i
    Find candidate overlapping predictions: $C \leftarrow \text{unique}(\mathbf{Y}_{\text{pred}}[\mathbf{M}_g]) \setminus \{0\}$
    $j^* \leftarrow 0$, BestIoU $\leftarrow 0.0$
    for $j \in C$ **do**
        $\mathbf{M}_p \leftarrow (\mathbf{Y}_{\text{pred}} == j)$
        IoU $\leftarrow \frac{\sum(\mathbf{M}_g \land \mathbf{M}_p)}{\sum(\mathbf{M}_g \lor \mathbf{M}_p)}$
        if IoU $>$ BestIoU **then**
            BestIoU $\leftarrow$ IoU, $j^* \leftarrow j$
        end if
    end for
    if $j^* > 0$ **then**
        $\mathbf{M}_{p^*} \leftarrow (\mathbf{Y}_{\text{pred}} == j^*)$
        IntersectionSum $+= \sum(\mathbf{M}_g \land \mathbf{M}_{p^*})$
        UnionSum $+= \sum(\mathbf{M}_g \lor \mathbf{M}_{p^*})$
        $\mathcal{U}_{\text{pred}} \leftarrow \mathcal{U}_{\text{pred}} \setminus \{j^*\}$ ▷ Mark prediction as matched
    else
        UnionSum $+= \sum \mathbf{M}_g$ ▷ Unmatched ground truth penalty
    end if
end for
for $j \in \mathcal{U}_{\text{pred}}$ **do**
    UnionSum $+= \sum \mathbb{I}(\mathbf{Y}_{\text{pred}} == j)$ ▷ Unmatched over-segmentation penalty
end for
return $\frac{\text{IntersectionSum}}{\text{UnionSum}}$

Algorithm 15 Panoptic Quality (PQ) Metric Computation

Require: Predicted instance mask $\mathbf{Y}_{\text{pred}} \in \mathbb{N}_0^{H \times W}$, Ground truth instance mask $\mathbf{Y}_{\text{gt}} \in \mathbb{N}_0^{H \times W}$
Ensure: Panoptic Quality metric score $\mathrm{PQ} = \mathrm{SQ} \times \mathrm{RQ} \in [0, 1]$

Extract instances $\mathcal{G} \leftarrow \text{unique}(\mathbf{Y}_{\text{gt}}) \setminus \{0\} = \{G_1, \ldots, G_K\}, \quad \mathcal{P} \leftarrow \text{unique}(\mathbf{Y}_{\text{pred}}) \setminus \{0\} = \{P_1, \ldots, P_M\}$
if $|\mathcal{G}| == 0 \wedge |\mathcal{P}| == 0$ **then return** 1.0
end if
if $|\mathcal{G}| == 0 \vee |\mathcal{P}| == 0$ **then return** 0.0
end if
$TP \leftarrow 0, \quad \text{IoUSum} \leftarrow 0.0$
Matched prediction set $\mathcal{M}_{\text{pred}} \leftarrow \emptyset$
for $i \leftarrow 1$ **to** K **do**
    $\mathbf{M}_g \leftarrow (\mathbf{Y}_{\text{gt}} == G_i)$
    Candidates $C \leftarrow \text{unique}(\mathbf{Y}_{\text{pred}}[\mathbf{M}_g]) \setminus \{0\}$
    for $j \in C$ **do**
        if $j \in \mathcal{M}_{\text{pred}}$ **then**
            continue
        end if
        $\mathbf{M}_p \leftarrow (\mathbf{Y}_{\text{pred}} == j)$
        $\text{IoU} \leftarrow \frac{\sum(\mathbf{M}_g \wedge \mathbf{M}_p)}{\sum(\mathbf{M}_g \vee \mathbf{M}_p)}$
        if $\text{IoU} > 0.5$ **then**
            $TP \mathrel{+}= 1$
            $\text{IoUSum} \mathrel{+}= \text{IoU}$
            $\mathcal{M}_{\text{pred}} \leftarrow \mathcal{M}_{\text{pred}} \cup \{j\}$
            break ▷ By uniqueness theorem, at most one prediction satisfies IoU > 0.5
        end if
    end for
end for
$FP \leftarrow |\mathcal{P}| - TP$ ▷ Unmatched predictions
$FN \leftarrow |\mathcal{G}| - TP$ ▷ Unmatched ground truths
$\text{SQ} \leftarrow \begin{cases} \frac{\text{IoUSum}}{TP}, & \text{if } TP > 0 \\ 0.0, & \text{otherwise} \end{cases}$ ▷ Segmentation Quality
$\text{RQ} \leftarrow \frac{TP}{TP + 0.5 \cdot FP + 0.5 \cdot FN}$ ▷ Recognition Quality
return $\text{SQ} \cdot \text{RQ}$

7. ALGORITHMIC COMPLEXITY ANALYSIS

The dual-stream formulation introduces negligible overhead during training and inference:

(a) **LLWT Computational Invariance**: Unlike 2D spatial Discrete Wavelet Transforms that incur $O(K \cdot H \cdot W)$ filter-bank convolutions, the Lifting Scheme splits features via subsampling ($O(1)$) and evaluates depthwise separable prediction filters over downsampled resolutions $\frac{H}{2} \times \frac{W}{2}$, reducing floating point operations by $4\times$.

(b) **BiCA Complexity**: Standard self-attention incurs quadratic spatial complexity $O(N^2 \cdot C)$ with $N = HW$. Channel cross-attention operates on $O(C^2 \cdot N)$ with $C \leq 1024 \ll N = 65,536$, ensuring bounded GPU memory footprints under 16 GB during training.

(c) **Inference Speed**: With sliding window inference (stride 192, 25% overlap) and vectorized watershed post-processing, an entire 1000×1000 tissue biopsy from MoNuSeg is segmented into verified cellular instances in under 2.8 seconds on an NVIDIA T4 GPU.

8. REFERENCES